

Supplementary Material for Backbone-Modular Neuro-Symbolic Post-Correction for OCT Denoising

Anonymous Author(s)¹

¹¹ Anonymous Affiliation

Correspondence withheld for double blind review.

This document holds the material that does not fit inside the page limit of the main paper. It adds nothing to the claims. It gives the closed form of every clinical property, the proof of the second theorem, the training algorithm, the full list of constants, the visual comparison for all four backbones, and a note on a defect found and repaired during revision.

S1. Closed form of the six clinical properties

Table S1 gives the score and the failure map of every property, and the expanded definitions of every subterm follow. All quantities are computed from the noisy input and the backbone output alone. No clean reference is used at any point.

Expanded subterms in the property table of the main paper. P_1 . $c_{\text{cont}} = \langle \text{erode}(\text{dilate}(b_e)), b_e \rangle / (\|b_e\|_1 + \varepsilon)$, $c_{\text{effr}} = \exp(-5|r_e - 0.2|)$ with $r_e = \|\mathbf{1}[e(x) > 1.5\bar{e}]\|_1 / (\|\mathbf{1}[e(x) < 0.5\bar{e}]\|_1 + \varepsilon)$, and $c_{\text{ang}} = \exp(-2 \text{mean}(\sigma_5(\theta(x))))$. P_2 . $s_{\text{range}} = \text{sigm}(10(\Delta_r - 0.3))$, $\Delta_r = \max(x) - \min(x)$. $s_{\text{cnr}} = \text{sigm}(15(\bar{\sigma} - 0.05))$, $\bar{\sigma} = \text{mean}(\sigma_9(x))$. $s_{\text{var}} = \max(0.2, \text{sigm}(-20(v_\sigma - 0.02) + 2))$, $v_\sigma = \text{Var}(\sigma_9(x))$. $s_{\text{cov}} = \text{sigm}(5(\gamma - 0.4))$, $\gamma = \text{mean}(\mathbf{1}[\sigma_9(x) > 0.3\bar{\sigma}])$. P_3 . $F = \mathbf{1}[e(x) < 0.3\bar{e}]$, $q_{\text{var}} = 1 - \langle F, \sigma_5(x) / (\sigma_5(y) + \varepsilon) \rangle / (\|F\|_1 + \varepsilon)$, and $q_{\text{res}} = \exp(-5 \text{mean}|y - x|)$. P_4 . s_{self} is the average Pearson correlation between x and shifts $(0, 2)$, $(2, 0)$, and $(2, 2)$. $s_{\text{reg}} = \exp(-12 \text{mean}(e(e(x))))$. $s_{\text{vert}} = \text{clip}(\max|\nabla_z v(x)| / (8 \text{mean}|\nabla_z v(x)|), 0, 1)$. P_5 . $z = \log(y + \varepsilon) - \log(x + \varepsilon)$, $cv = \sigma_{15}(|z|) / (\mu_{15}(|z|) + \varepsilon)$, and $cv^*(d)$, where d is the axial depth index, is the depth-wise Gamma-K target coefficient of variation used by the implementation. P_6 . $s_{\text{clr}} = \text{mean}(\text{clip}(\max|\nabla_z v(x)| / (10 \text{mean}|\nabla_z v(x)|), 0, 1))$, $s_{\text{dep}} = \text{clip}(20 \text{Var}(v(x)), 0, 1)$, and $s_{\text{smo}} = 1 - \text{clip}(10 \text{mean}(\text{Var}_z(\text{mean}_w e(x))), 0, 1)$.

These are the implemented predicates. The maps $\mathbf{m}_i$ are the spatial quantities consumed by the negotiator and potential head, while the scores s_i drive the rule activations. The released code computes exactly these quantities. The relevant class is named `EnhancedGTFreePredicates` and the speckle property is computed by `PhysicsAccurateSpecklePredicate`.

S2. Proofs

A. Proof of the allocation stability result

This is the proof of Theorem 1 of the main paper (Allocation stability), repeated here in full. Write $s_k = -1$ for $k \in \{1, 5\}$ and $s_k = +1$ otherwise, so that the allocation equation of the main paper reads $a(\mathbf{r}) = b + \sum_k s_k C_k w_k r_k$ with b independent of $\mathbf{r}$, where $\mathbf{r} = (r_1, \dots, r_5) \in [0, 1]^5$ is the vector of rule truth values at one pixel, C_k are the fixed budgets, and $w_k = \sigma(\theta_k)$ are the learned rule weights.

Proof:

Subtracting two such expressions gives

$$a(\mathbf{r}) - a(\mathbf{r}') = \sum_k s_k C_k w_k (r_k - r'_k).$$

Taking absolute values and applying the triangle inequality,

$$|a(\mathbf{r}) - a(\mathbf{r}')| \leq \sum_k C_k w_k |r_k - r'_k| \leq \left(\sum_k C_k w_k \right) \|\mathbf{r} - \mathbf{r}'\|_\infty,$$

which is the stability bound of Definition 1 of the main paper with $L = \sum_k C_k w_k$ as claimed. The bound is attained by taking $r_k - r'_k = s_k t$ for any t small enough to keep both vectors in the unit cube, since then every term of the sum has the same sign. Because each C_k is positive and each $w_k = \sigma(\theta_k)$ lies strictly in $(0, 1)$, the sum is strictly less than $\sum_k C_k = 0.75$. ■

B. Proof of the contrast margin result

This is the proof of Theorem 2 of the main paper, repeated here in full. Write $\mu_{\mathcal{T}}$ and $\mu_{\mathcal{B}}$ for the mean over tissue and over background, and $\sigma_{\mathcal{B}}$ for the standard deviation over background. Let $\mathbf{b}$ be the backbone output, $\mathbf{q}$ the candidate, and $\hat{\mathbf{x}} = (1 - w)\mathbf{b} + w\mathbf{q}$ the output with $w \in (0, 1]$.

Theorem S1 (Contrast margin). Let $\eta_{\mathcal{T}} = \mu_{\mathcal{T}}(\mathbf{q}) - \mu_{\mathcal{T}}(\mathbf{b}) \geq 0$ and let $\eta_{\mathcal{B}} \geq \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b})$. If $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$ then

$$\text{CNR}(\hat{\mathbf{x}}) \geq \frac{(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})}{\sigma_{\mathcal{B}}(\mathbf{b})}. \quad (\text{S1})$$

TABLE S1. Closed form of the six clinical properties. For each predicate, the table lists its clinical role, image-level score, and pixel-level failure map. larger $\mathbf{m}_i$ means stronger evidence that the corresponding property is locally unsatisfied. The expanded subterm definitions follow immediately below. Here $\bar{e} = \text{mean}(e(x))$, $b_e = \mathbf{1}[e(x) > \bar{e}]$, $v(z) = \text{mean}_w z$ is a depth profile, ρ_h is 1-pixel horizontal autocorrelation, and Var_k is local variance over a $k \times k$ window.

P_i	Clinical role	Aggregate score s_i	Failure map $\mathbf{m}_i$	τ_i
P_1	Edge continuity	$0.4 c_{\text{cont}} + 0.3 c_{\text{efr}} + 0.3 c_{\text{ang}}$	$(1 - \bar{e})(1 - b_e)$, $\bar{e} = e(x)/(e_{\text{max}} + \varepsilon)$	0.6
P_2	Local contrast recovery	$0.20 s_{\text{range}} + 0.35 s_{\text{cnr}} + 0.20 s_{\text{var}} + 0.25 s_{\text{cov}}$	$[\text{ReLU}(\bar{\sigma} - \sigma_9(x))/(\bar{\sigma} + \varepsilon) + 0.3 \mathbf{1}[\sigma_9(x) < 0.01]]_{[0,1]}$	0.5
P_3	Flat-region smoothness	$0.6 q_{\text{var}} + 0.4 q_{\text{res}}$	$F \odot \sigma_5(x)/(\max \sigma_5(x) + \varepsilon)$	0.75
P_4	Structural coherence	$0.4 s_{\text{self}} + 0.3 s_{\text{reg}} + 0.3 s_{\text{vert}}$	$e(e(x))/(\max e(e(x)) + \varepsilon)$	0.5
P_5	Speckle conformity	$0.7 \text{mean}(\exp(- cv - cv^* /0.15)) + 0.3 \exp(-5 \rho_h(z))$	$1 - \exp(- cv - cv^* /0.15)$	0.3
P_6	Layer visibility	$0.4 s_{\text{clr}} + 0.3 s_{\text{dep}} + 0.3 s_{\text{smo}}$	$\text{Var}_7(e(x))/(\max \text{Var}_7(e(x)) + \varepsilon)$	0.7

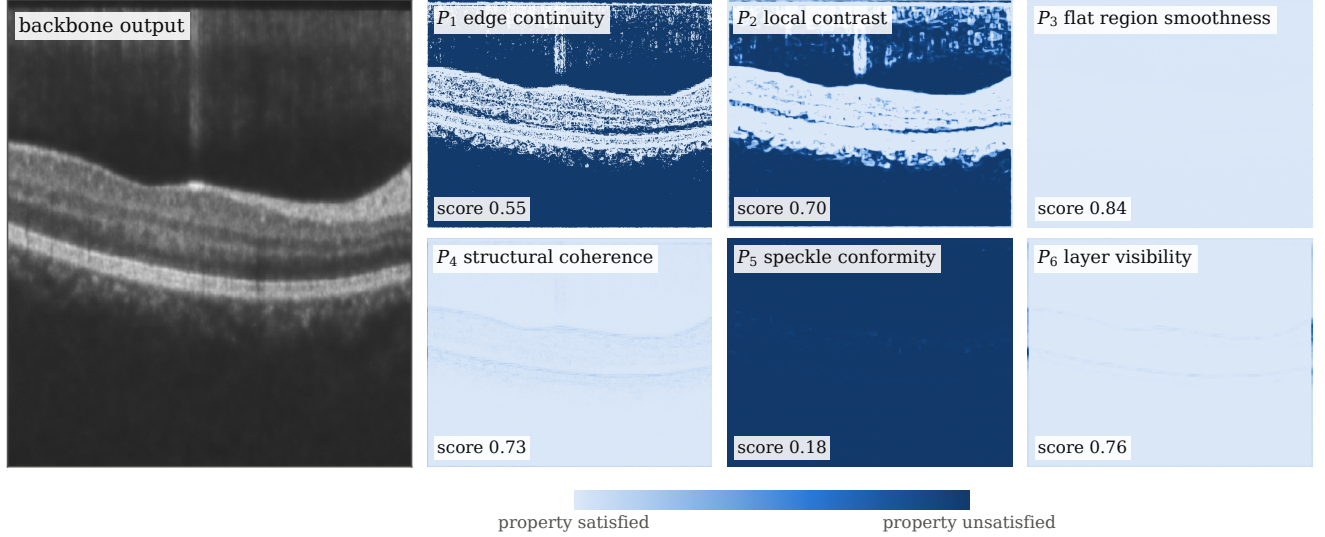


FIGURE S1. The six clinical properties on one backbone output. Each panel shows where that property is locally unsatisfied, on a shared scale, with the image level score in the corner. Edge continuity, local contrast and speckle conformity carry most of the evidence on this image, while the other three are close to satisfied everywhere.

Proof:

The mean is linear, so over any fixed set of pixels the mean of a convex combination is the same convex combination of the means. Applying this over the tissue set and over the background set,

$$\begin{aligned}\mu_{\mathcal{T}}(\hat{\mathbf{x}}) &= (1 - w) \mu_{\mathcal{T}}(\mathbf{b}) + w \mu_{\mathcal{T}}(\mathbf{q}) = \mu_{\mathcal{T}}(\mathbf{b}) + w \eta_{\mathcal{T}}, \\ \mu_{\mathcal{B}}(\hat{\mathbf{x}}) &= \mu_{\mathcal{B}}(\mathbf{b}) + w \eta'_{\mathcal{B}},\end{aligned}$$

where $\eta'_{\mathcal{B}} = \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b}) \leq \eta_{\mathcal{B}}$ by hypothesis. Subtracting,

$$\mu_{\mathcal{T}}(\hat{\mathbf{x}}) - \mu_{\mathcal{B}}(\hat{\mathbf{x}}) = (\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta'_{\mathcal{B}}) \geq (\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}}),$$

which lower bounds the numerator.

For the denominator, every background pixel of $\hat{\mathbf{x}}$ is the same convex combination of the corresponding pixels of $\mathbf{b}$ and $\mathbf{q}$. Treating the background pixel values as a random variable and applying Minkowski's inequality to the centred variables,

$$\sigma_{\mathcal{B}}(\hat{\mathbf{x}}) \leq (1 - w) \sigma_{\mathcal{B}}(\mathbf{b}) + w \sigma_{\mathcal{B}}(\mathbf{q}) \leq (1 - w) \sigma_{\mathcal{B}}(\mathbf{b}) + w \sigma_{\mathcal{B}}(\mathbf{b}) = \sigma_{\mathcal{B}}(\mathbf{b}),$$

using the hypothesis $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$ in the second step. Dividing the lower bound on the numerator by the upper bound on the denominator gives the claim. Since $w > 0$ in every branch of the blending rule, the bound is at least the baseline contrast whenever $\eta_{\mathcal{T}} > \eta_{\mathcal{B}}$. ■

The results section of the main paper reports how often each of the three hypotheses holds on the test set, so the reader can see when the guarantee is active rather than assuming it always is.

S3. Training procedure and the full list of constants

A. Algorithm

Algorithm 1 states the procedure. The backbone is frozen throughout and receives no gradient. The first stage activates only the clinical terms, so the correctors learn a useful direction. The second stage activates the fidelity so they learn how far to go.

Algorithm 1 Training the wrapper.

Require: frozen backbone f_θ , paired training set, number of epochs E

- 1: initialise the wrapper parameters, set each property threshold to its literature value
- 2: for $e = 1$ to E do
- 3: for each batch of crops do
- 4: compute the backbone output, with no gradient to θ
- 5: compute the cooperation map from the backbone features
- 6: compute the six property scores and failure maps
- 7: compute the gain, edge and strength proposals
- 8: compute the five rule truth values
- 9: compute the allocation
- 10: compose the candidate, then evaluate the objective on it
- 11: update the wrapper parameters with AdamW
- 12: end for
- 13: if e reaches the midpoint then activate the fidelity term
- 14: end if
- 15: end for

B. Schedule and optimiser settings

We use AdamW with decoupled weight decay of 10^{-4} , random crops of 96×96 pixels, and a batch of sixteen crops, reduced to eight for DnCNN and SwinIR, the two with the largest memory footprint. Three learning rates are used, a lower one for the corrector networks and higher ones for the strength predictor and the rule layer, because the last two have far fewer parameters and therefore need larger steps to move at all. The switch from the clinical only stage to the stage that also includes fidelity happens at the midpoint of the schedule, matching the epoch check in Algorithm 1.

C. Constants that carry a design decision

Table S2 lists the constants that carry a design decision, where each acts, and how it was chosen. A constant marked measured was set from a property of the data. A constant marked selected was chosen on the validation split and never on the test split. A constant marked fixed is a structural choice that was not tuned.

The networks use the default initialisation of their layers. Two exceptions are deliberate. The bias of the brightness head starts at a small positive value so that the multiplicative branch begins from a mild brightening rather than from zero, which avoids a flat region of the loss at the start of training. The threshold of each property is declared as a trainable parameter and set to the value reported in the clinical literature for that

TABLE S2. Every constant in the method, where it acts and how it was chosen. Fixed means a structural choice that was not tuned. Set in development means it was chosen once on the validation split while the method was being built and never revisited for this revision. Selected in the revision means it entered the grid search of the revision, on the validation split and never on the test split. Learned means it is a parameter of the model fitted by the objective. No constant was ever chosen on the test split.

Constant	Value	Acts on
$\alpha_{\max}$	0.15	scale of the raw gain
gain bound	± 0.20	Eq. (5) of the main paper
gate percentile	0.15	Eq. (6) of the main paper
κ	10	Eq. (3) of the main paper
$B_{\min}, B_{\max}$	0.25, 0.50	base of Eq. (4) of the main paper
$C_1 \dots C_5$	0.15, 0.25, 0.15, 0.10, 0.10	rule votes
$\tau_1 \dots \tau_6$	0.6, 0.5, 0.75, 0.5, 0.3, 0.7	Eq. (3) of the main paper
$\beta, \theta_1 \dots \theta_5$	learned	Eq. (4) of the main paper
energy tolerance	0.10	constraint one
Pareto tolerance	0.10	constraint two
change bound	0.2	constraint three
background blend	0.5	background smoothing
learning rates	2×10^{-4} , 5×10^{-4} , 3×10^{-4}	training
fidelity dead zone d	0.6 or 0.9 dB	Eq. (14) of the main paper
gate rule	percentile or tissue bounded	Eq. (6) of the main paper

TABLE S3. Structural constants. None of these was tuned.

Constant	Value	Role
gate steepness	20	sharpness of the background gate
failure sharpness	10	sharpness of the failure conversion
smoothing window	3×3	smoothing of the background term
erosion window	5×5	keeps background away from tissue
crop size	96×96	training crops
batch size	16, or 8 for DnCNN and SwinIR	training batch
edge bound	± 0.08	nominal bound of the multiplicative branch
quarter resolution	$H/4 \times W/4$	the gain network at this scale

property. It receives no gradient under the objective of the main paper, and in every selected checkpoint all six sit at their literature values to four decimals, so they are fixed in effect and the main paper reports them as fixed.

D. Structural constants

Table S3 lists the remaining constants, which are structural and were not tuned.

TABLE S4. The gate rule and fidelity dead zone selected for each backbone. The radius column applies only to the tissue bounded gate.

Backbone	Gate rule	Radius	Dead zone
NAFNet	tissue bounded	41	0.6
DnCNN	percentile	not applicable	0.9
KBNet	percentile	not applicable	0.9
SwinIR	percentile	not applicable	0.9

TABLE S5. Feature tensor read by the cooperation head for each backbone. All heads share the same two layer design and differ only in the number of input channels.

Backbone	Feature tensor	Channels
NAFNet	First encoder stage	40
KBNet	First encoder stage	32
DnCNN	First convolution	228
SwinIR	Shallow feature convolution	138

E. Selected gate rule and dead zone per backbone

Table S4 lists, for each backbone, the gate rule chosen by the search described in the training section of the main paper, the radius at which a tissue bounded gate applies, and the selected fidelity dead zone.

F. What the paired data is used for

Training uses paired data. The clean reference enters the fidelity term, the clinical term and the supervision of the edge branch. Testing uses no clean image anywhere in the correction path. The six properties are computed from the noisy input and the backbone output only, which is what allows them to run on a scanner the method has never seen. The clean reference is used to score results, as in any denoising study, and never to produce them.

S4. Visual comparison for all four backbones

Fig. S2 repeats the visual comparison of the main paper for all four backbones at full resolution. The layout is the same. Every cell is a zoomed region chosen automatically as the window inside tissue where the wrapper changed the NAFNet output most, because a change too small to see at full scale is the normal case and the reader should be shown the largest one rather than an average one. The last column shows the change the wrapper applied, on a scale symmetric about zero, so the reader can see where the correction acted rather than inferring it.

S5. A defect found during revision, and its repair

We record this because the released code contains a switch that restores the earlier behaviour, and a reader who finds that switch deserves to know what it is for.

A. What was wrong

In the version of this work that was first submitted, the allocation was formed by adding the rule contributions to a base level and then clipping the result to the unit interval. The positive contributions could reach 0.884 and 0.762 on top of a base of 0.818, so the value arriving at the clip could be as large as 2.46. The clip was therefore active at every pixel of every image and the allocation was the constant one.

Two consequences follow. The rule layer had no influence on any output, since multiplying by one changes nothing. And the derivative of a clip at its bound is zero, so no gradient reached the rule parameters and they stayed at their initial values for the whole of training. We confirmed both directly, by reading the allocation map out of the trained model and by comparing every rule parameter in the saved weights against its initialiser after 2082 optimiser steps.

B. The repair

The allocation is now a bounded weighted vote. Writing $r_k \in [0, 1]$ for the truth value of rule k and $w_k \in (0, 1)$ for its learned weight,

$$a = B_{\min} + (B_{\max} - B_{\min}) \sigma(\beta) - C_1 w_1 r_1 + C_2 w_2 r_2 + C_3 w_3 r_3 + C_4 w_4 r_4 \quad (\text{S2})$$

with $B_{\min} = 0.25$, $B_{\max} = 0.50$ and budgets $C_1 = 0.15$, $C_2 = 0.25$, $C_3 = 0.15$, $C_4 = 0.10$ and $C_5 = 0.10$. The budgets satisfy

$$B_{\max} + C_2 + C_3 + C_4 = 1, \quad B_{\min} - C_1 - C_5 = 0, \quad (\text{S3})$$

so the value lies in the unit interval before any clipping, for every input and every parameter setting. Clipping is therefore never active, saturation is impossible rather than unlikely, and the gradient reaches every rule parameter.

The repair also tightens the theory. Because the allocation is affine in the rule truth values with signed coefficients and no clipping intervenes, the Lipschitz constant of Theorem 1 of the main paper is attained rather than merely an upper bound.

This is also what the trained model shows and not only a property of the formula. Measured on the released weights, the allocation now spans 0.556 to 0.793 with mean 0.726 and standard deviation 0.050, and 0.00 percent of pixels sit at either end of the clamp, against the whole image before the repair, direct confirmation that no pixel saturates in practice.

C. Reproducing the earlier behaviour

Setting the environment variable `LEGACY_SATURATING_ALLOCATION` to 1 restores the original computation exactly. We verified this against a copy of the pre revision source on twenty random inputs, comparing the allocation maps for equality and the rule traces for agreement. Both

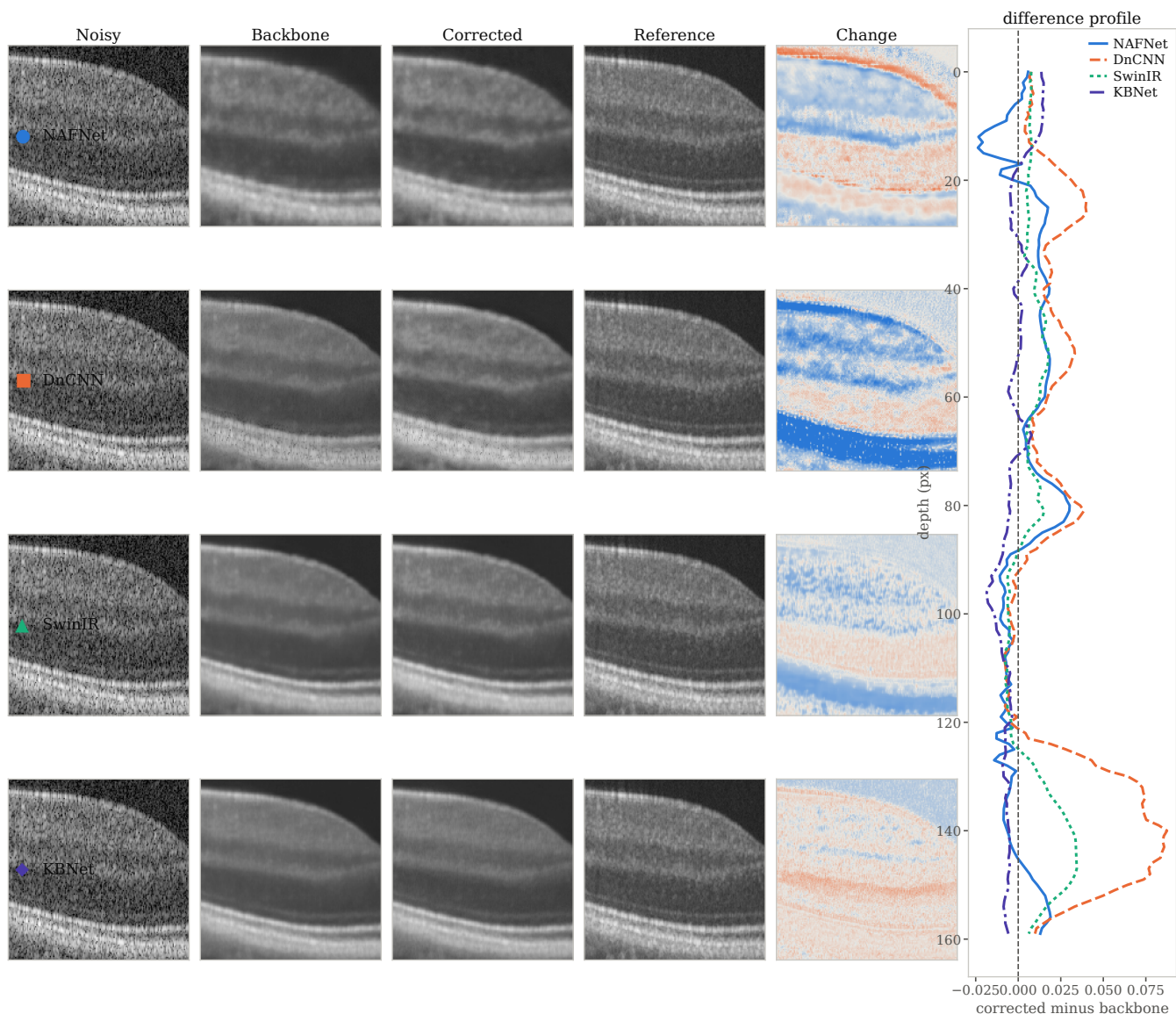


FIGURE S2. Visual comparison on PKU37 for all four backbones. Columns are the noisy input, the backbone output, the corrected output, the clean reference, and the change applied. Orange means the wrapper lowered the value and blue means it raised it. The rightmost panel is the difference profile across depth for all four backbones over the same crop.

matched. The earlier numbers can therefore be recovered from the released code, which is why the switch exists.

S6. Leave one out procedure for the clinical properties

The results section of the main paper reports the contribution of each clinical property. The procedure is a leave one out ablation. At test time, one property at a time is switched off by setting its score to the satisfied value and its failure map to zero, and the whole test set is re-evaluated with that property held out. This is the same kind of one at a time removal used for the rule constants and for the components of the method, applied here to the six clinical properties instead.

S7. Full per cell spreads

The results table of the main paper prints each clinical cell as a mean, with the robustness test folded into the choice of bold, and drops the per seed spread to keep the table one page. Table S6 here restores that spread for every cell, four backbones by three blocks by six measures, each as mean plus or minus one standard deviation across three seeds, generated directly from the same result files.

S8. The cooperation map, in full

The main paper reports that the map produced by the head reading the backbone features does not predict the error of that backbone. This section gives the rest of the analysis.

TABLE S6. Every clinical cell of the merged results table, with its full mean plus or minus one seed standard deviation restored. Each cell here is the same one that table shows as a mean alone, where bold marks a mean that exceeds its spread.

Backbone	ΔCNR	ΔTCI	ΔEPI	ΔBS	ΔENL	ΔSNR
PKU37 test set, 173 images						
NAFNet	$+3.49 \pm 0.19$	$+1.91 \pm 0.51$	$+1.20 \pm 0.02$	$+3.89 \pm 0.45$	$+23.74 \pm 0.38$	$+11.10 \pm 0.31$
DnCNN	$+11.30 \pm 0.52$	$+0.99 \pm 1.62$	$+4.14 \pm 0.16$	-1.44 ± 0.87	$+57.14 \pm 7.63$	$+22.95 \pm 2.28$
SwinIR	$+7.33 \pm 0.81$	$+11.45 \pm 0.86$	$+2.68 \pm 0.16$	$+1.82 \pm 0.37$	$+28.32 \pm 1.30$	$+11.54 \pm 0.26$
KBNet	$+6.93 \pm 0.98$	$+5.92 \pm 1.81$	$+0.00 \pm 0.07$	$+2.58 \pm 1.35$	$+25.30 \pm 2.17$	$+7.95 \pm 0.36$
Duke17, no adaptation						
NAFNet	$+7.51 \pm 0.21$	$+4.00 \pm 1.08$	$+1.37 \pm 0.07$	$+4.15 \pm 0.16$	$+21.35 \pm 1.78$	$+5.43 \pm 0.80$
DnCNN	$+12.19 \pm 0.72$	-1.47 ± 3.15	$+3.49 \pm 0.15$	-2.24 ± 1.55	$+39.42 \pm 5.44$	$+13.55 \pm 1.35$
SwinIR	$+8.68 \pm 1.12$	$+13.26 \pm 0.91$	$+2.69 \pm 0.24$	$+1.90 \pm 1.00$	$+29.46 \pm 0.66$	$+11.60 \pm 0.29$
KBNet	$+5.34 \pm 0.87$	$+7.19 \pm 1.57$	-0.50 ± 0.19	$+1.61 \pm 0.09$	$+7.49 \pm 3.76$	-1.81 ± 1.80
Duke2013, no adaptation						
NAFNet	$+7.37 \pm 0.06$	$+1.41 \pm 0.87$	$+0.85 \pm 0.04$	$+0.51 \pm 0.16$	$+28.80 \pm 2.25$	$+9.29 \pm 1.25$
DnCNN	$+12.73 \pm 1.41$	-2.21 ± 2.71	$+2.99 \pm 0.25$	-5.85 ± 0.87	$+49.91 \pm 7.29$	$+17.59 \pm 2.04$
SwinIR	$+9.26 \pm 1.20$	$+11.86 \pm 1.10$	$+2.55 \pm 0.20$	$+2.30 \pm 1.69$	$+30.90 \pm 1.02$	$+13.31 \pm 0.20$
KBNet	$+7.16 \pm 0.94$	$+4.83 \pm 1.32$	-0.35 ± 0.09	$+3.06 \pm 1.03$	$+8.70 \pm 4.55$	-0.90 ± 2.14

Two measures were computed over the 173 test images. The rank correlation between the map and the absolute error of the backbone output with respect to the clean reference is -0.258 . The sparsification error, defined as the area between the curve obtained by removing the pixels the map calls least reliable and the curve obtained by removing the pixels that are actually worst, is 0.775 . A perfect ordering scores zero on the second measure and one on the first.

The reason is structural rather than a matter of tuning. The head is trained only through the cooperation term of the objective, which rewards agreement between the map and the estimated correction potential. That potential is itself a learned quantity. A map supervised only against another learned map has no path by which the error of the backbone could reach it, so there is no reason for it to acquire the meaning the original name implied.

Repairing this would require a term that regresses the map on the measured backbone error. Such a term would compete with the cooperation term rather than complement it, because one pulls the map toward the error of the backbone and the other pulls it toward where the correctors want to act. The relative weight of the two would have to be selected, and every result in this paper would have to be recomputed. We therefore report the measurement and leave the repair as the clearest piece of future work this revision has identified.

S9. The adapted transfer protocol of the original submission

The cross dataset numbers of the original submission were not produced by the no adaptation protocol the

text described. The script that produced them resumed the corrector trained on PKU37 and trained it further on the target dataset in a leave one subject out design. We rerun that protocol here with the selected NAFNet checkpoint so that the gap between the two protocols can be read directly, and we describe it exactly.

A. Protocol

Each fold holds out one subject, seventeen of Duke2013 or fifteen of Duke17 subjects form the training set, and the held out subject is evaluated once. Training resumes from the PKU37 checkpoint of the tissue bounded gate setting at seed 0 and runs for fifteen epochs with batch size two. The backbone weights and its normalisation statistics are frozen. The wrapper and the cooperation head that reads the backbone features are trainable. The learning rates are 2×10^{-5} , 1×10^{-4} and 6×10^{-5} for the corrector, potential and negotiator, five to ten times below the PKU37 schedule of Table S2, the fidelity term enters at epoch five, and an L2 penalty of weight 0.1 holds every trainable weight near its PKU37 value, which is elastic weight consolidation without the Fisher weighting. A background correction variance penalty of weight 10 is also active. The checkpoint kept for each fold is the epoch with the highest validation score, and the validation image of a fold is its held out subject. The adapted numbers are therefore selected on the subject they are evaluated on and should be read as an upper bound for this protocol. The same was true of the original submission.

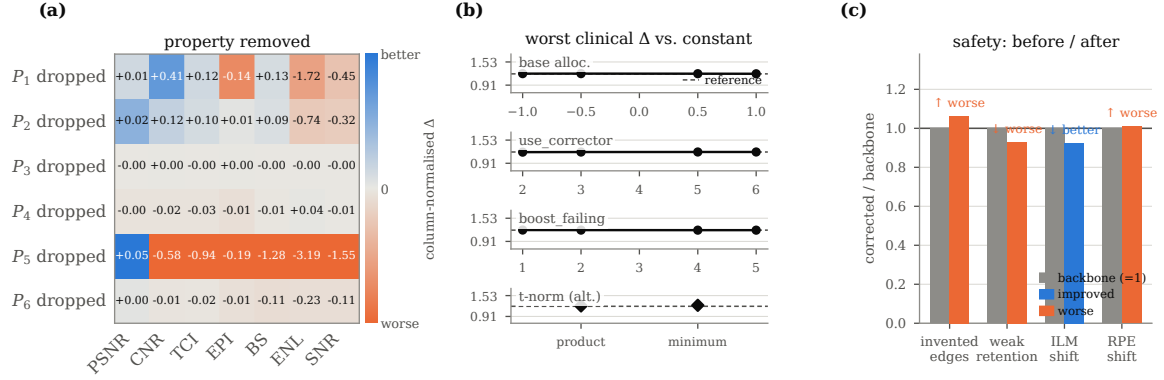


FIGURE S3. The three studies of the main paper drawn together. Left, removing one clinical property at a time, in points of the measure. Only P_5 , P_1 and P_2 move any measure by more than a quarter point, so P_3 , P_4 and P_6 are close to inert here. Middle, the fourteen variations of the rule constants, none of which turns a clinical measure negative, with the base allocation level the only visible effect. Right, the safety analysis, each of the invented edge rate, weak structure retention and the two boundary shifts shown as the ratio of corrected to backbone value, one meaning unchanged, the arrow giving the direction and the word whether it is better or worse for that measure. Three ratios sit on the worse side, though the retinal pigment epithelium shift moves one percent and is not significant. All panels are the selected NAFNet checkpoint on the same 173 test images.

TABLE S7. NAFNet, adapted protocol over one leave one subject out fold per subject, against the no adaptation protocol of Table 3 of the main paper on the same checkpoint. Clinical rows are percentage change from backbone output to corrected output. The spread is the standard deviation across subjects, the count is the number of subjects on which the measure improved, and p is a two sided Wilcoxon signed rank test on the paired per subject values.

Measure	No adaptation	Adapted	Improved
Duke17, 16 subjects			
Δ PSNR (dB)	-0.17	-0.23 ± 0.44	6 of 16
Δ CNR	+7.51	$+10.68 \pm 1.55$	16 of 16
Δ TCI	+4.00	$+2.69 \pm 4.16$	12 of 16
Δ EPI	+1.37	$+1.49 \pm 0.78$	15 of 16
Δ BS	+4.15	$+4.13 \pm 5.18$	13 of 16
Δ ENL	+21.35	$+24.29 \pm 17.72$	15 of 16
Δ SNR	+5.43	$+2.95 \pm 8.97$	9 of 16
Duke2013, 18 subjects			
Δ PSNR (dB)	-0.35	-0.45 ± 0.47	4 of 18
Δ CNR	+7.37	$+15.43 \pm 4.09$	18 of 18
Δ TCI	+1.41	$+5.15 \pm 3.75$	17 of 18
Δ EPI	+0.85	$+0.90 \pm 0.89$	15 of 18
Δ BS	+0.51	$+2.96 \pm 2.97$	14 of 18
Δ ENL	+28.80	$+9.23 \pm 28.61$	8 of 18
Δ SNR	+9.29	-5.29 ± 13.97	3 of 18

subjects losing more than 1 dB, and a signal to noise ratio 2.5 and 14.6 points lower. The mean absolute correction rises from 0.0104 to 0.0121 on Duke17 and from 0.0118 to 0.0164 on Duke2013. The other measures do not agree between the two sets. On Duke17 the tissue contrast index falls from +4.0 to +2.7 percent while the speckle measure rises, and no measure falls below zero, although the signal to noise ratio is no longer significant. On Duke2013 the tissue contrast index, edge preservation and boundary sharpness all rise, the speckle measure changes on eight subjects one way and ten the other and is not significant, and the signal to noise ratio falls by 0.513 percent, a decrease on fifteen of eighteen subjects that is significant at the one percent level. We draw one conclusion. Adaptation on fifteen to seventeen images of the target scanner reliably improves one measure, contrast, which the frozen model already improves on both sets. It pays for that in fidelity and signal to noise ratio on both sets, its effect on the remaining measures changes direction between two scanners, and it does so from a checkpoint selected on the held out subject. The frozen model gives six positive measures on both sets with no fitting on the target data, and it is the protocol the main paper reports and the one we recommend.

B. Result

Table S7 gives both target sets, next to the no adaptation result of the same checkpoint from the main paper.

Two things hold on both scanners. Adaptation raises contrast, by 3.2 points on Duke17 and 8.1 on Duke2013, and every subject improves. Adaptation also costs fidelity and signal to noise ratio relative to the frozen model, a further 0.06 and 0.10 dB, with one and two